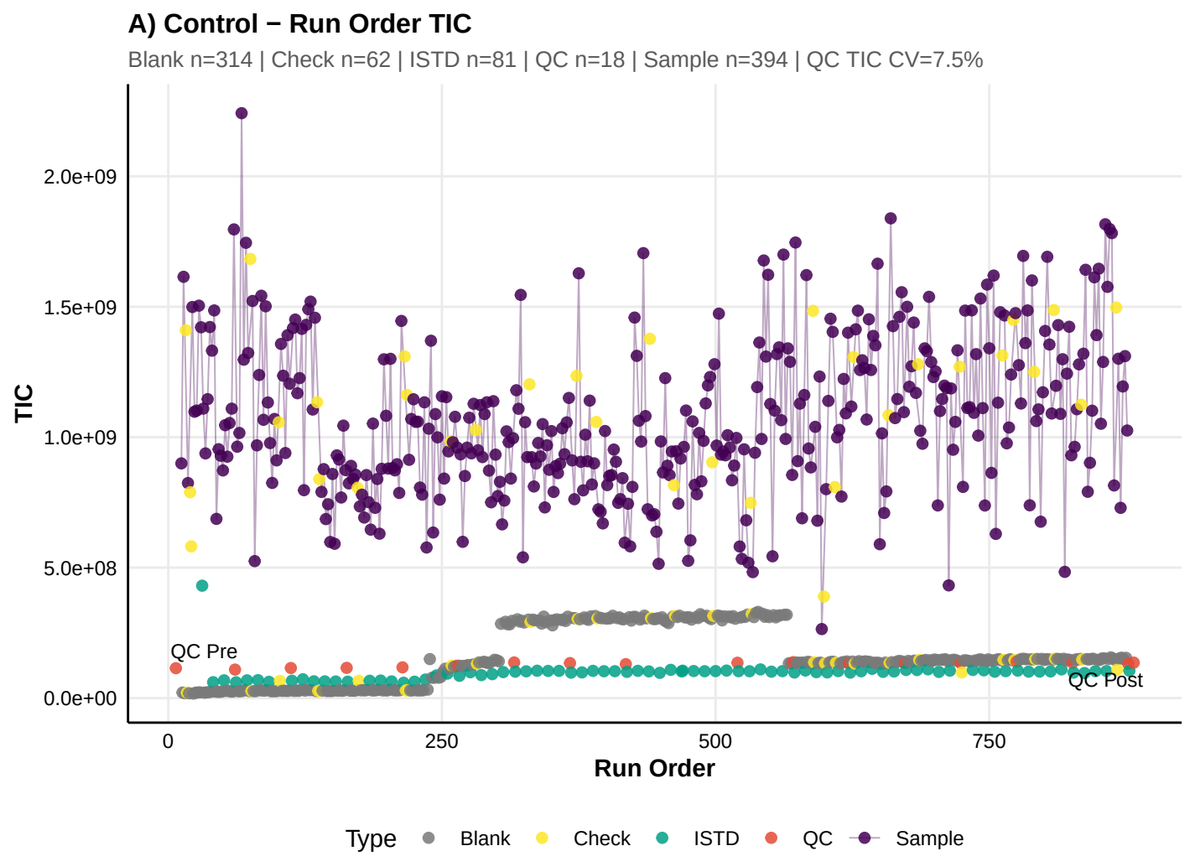
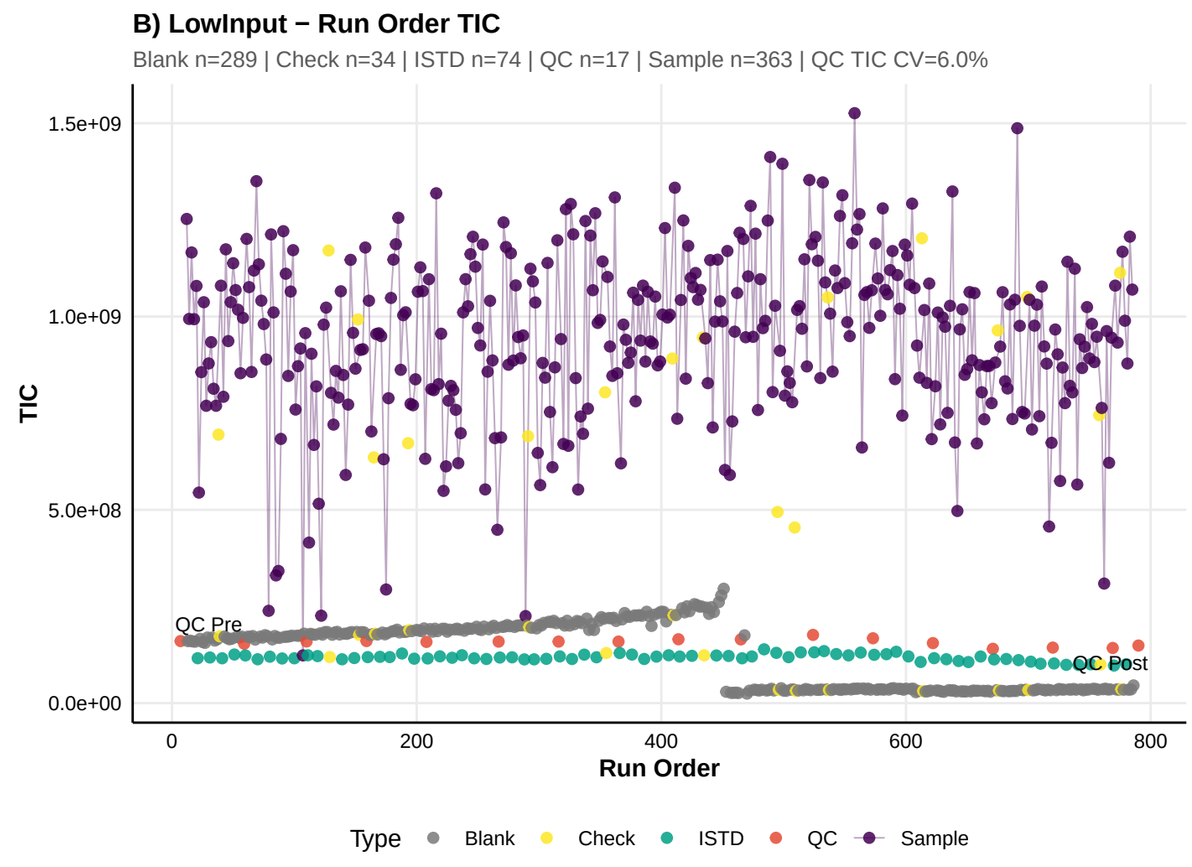
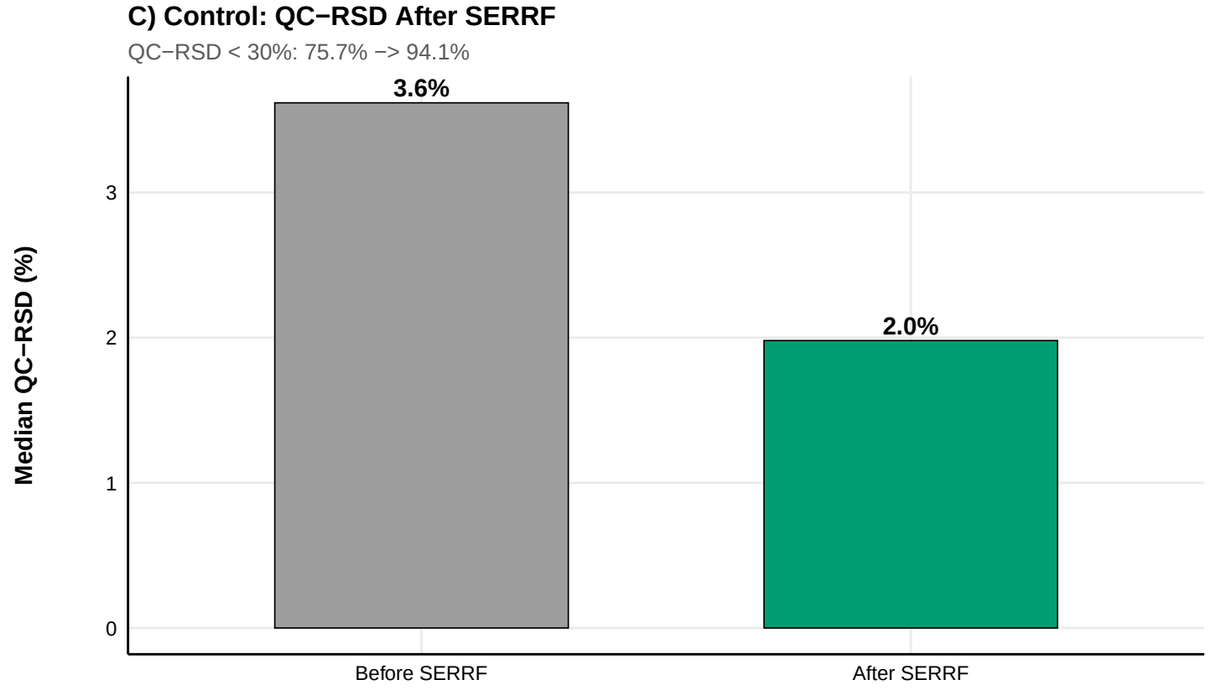
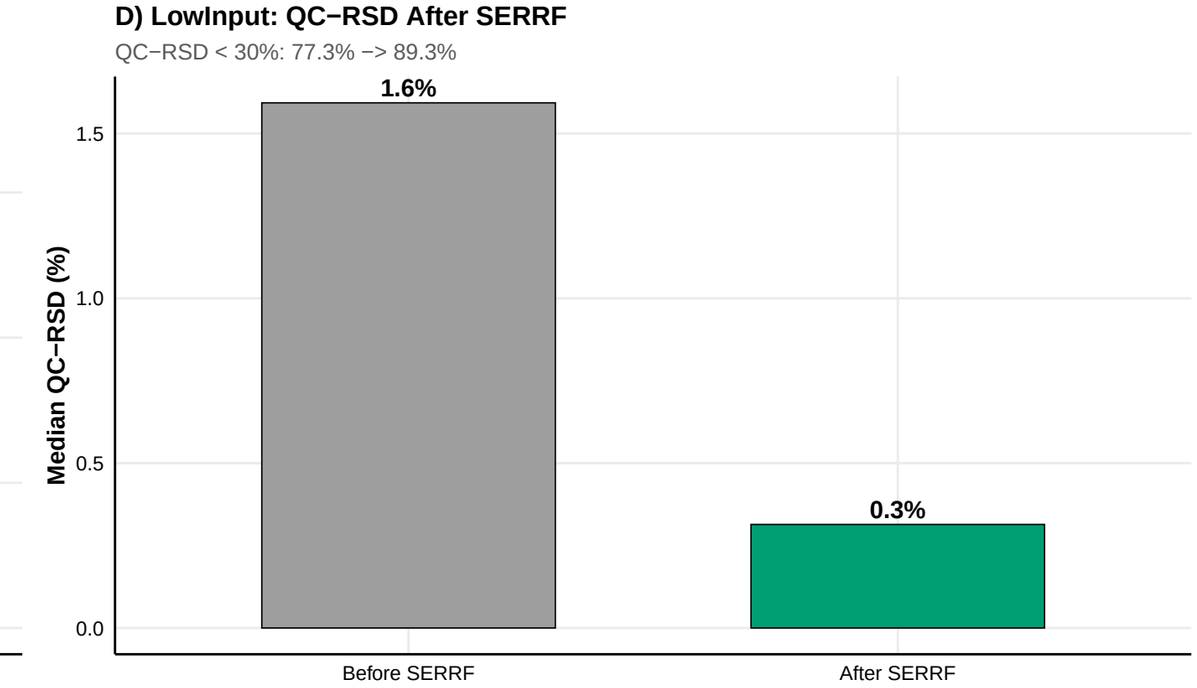
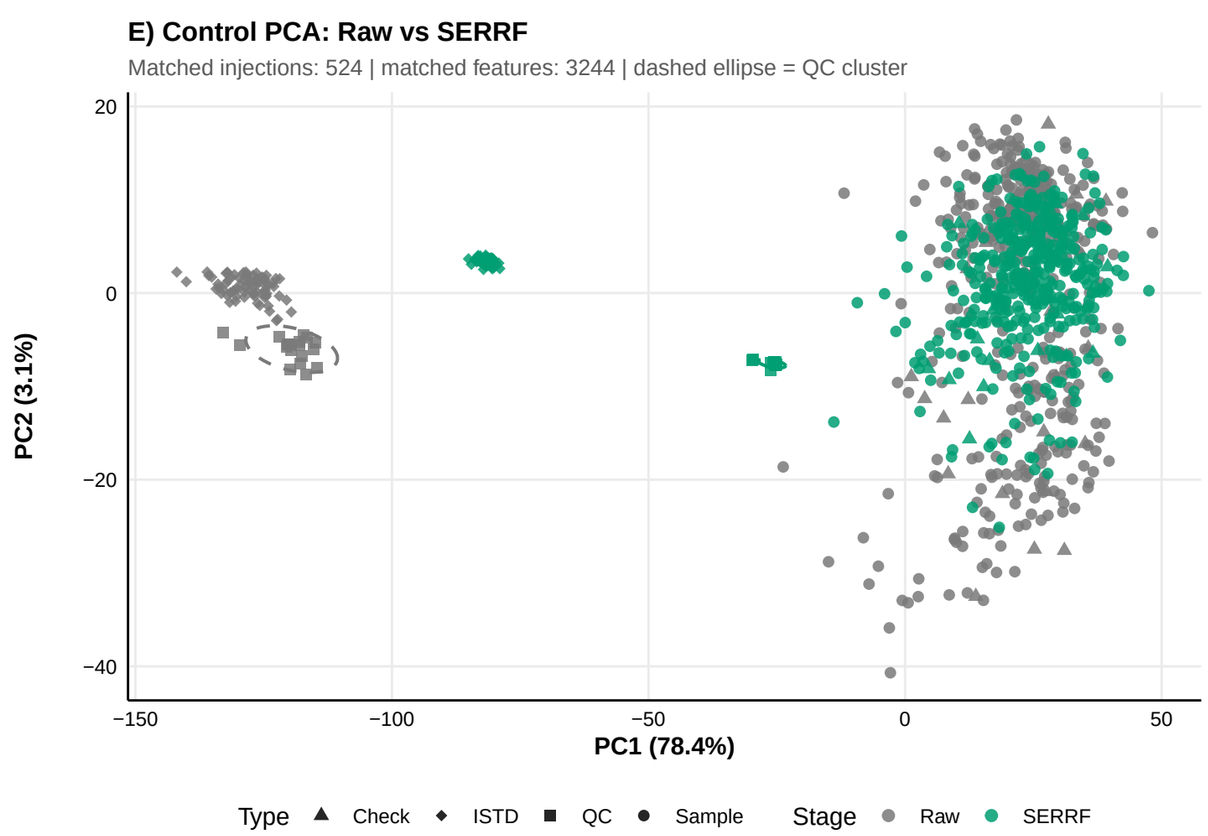
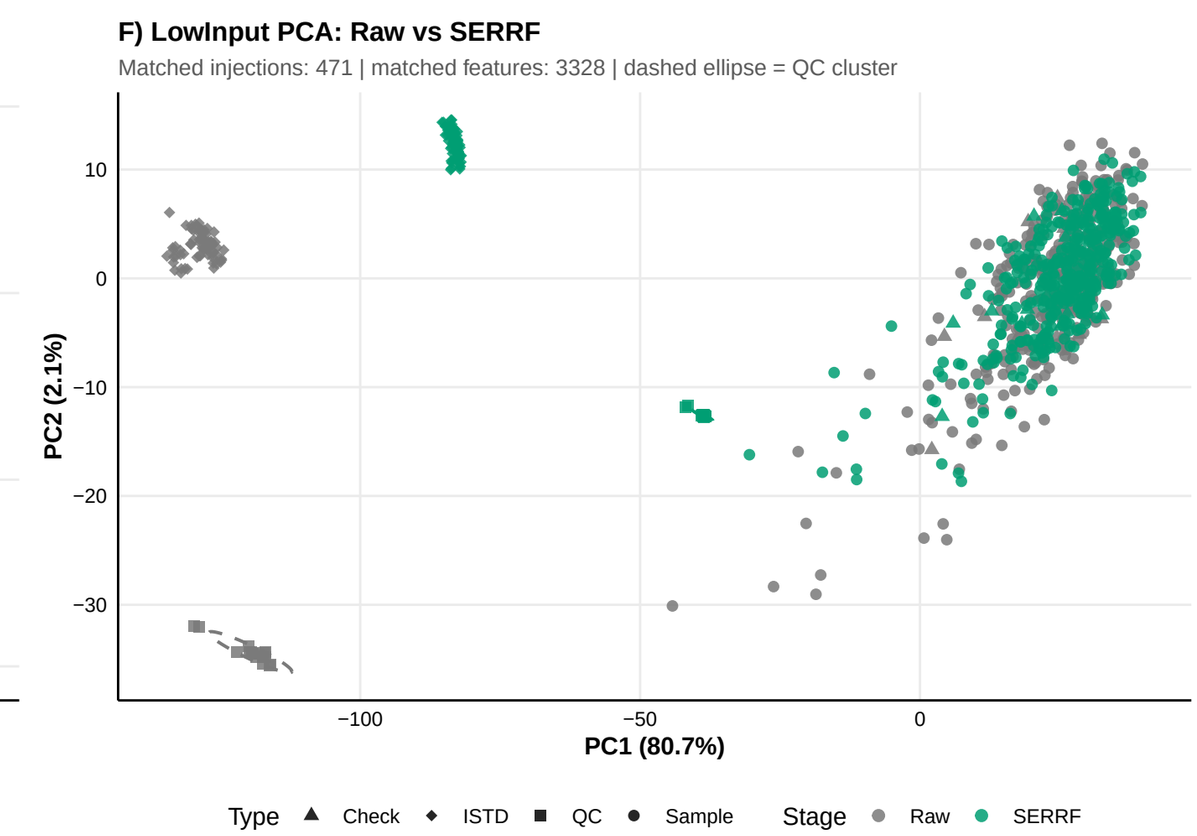

A) Control − Run Order TIC
Blank n=314 | Check n=62 | ISTD n=81 | QC n=18 | Sample n=394 | QC TIC CV=7.5%
TIC
Run Order
QC Pre
QC Post
Type Blank Check ISTD QC Sample

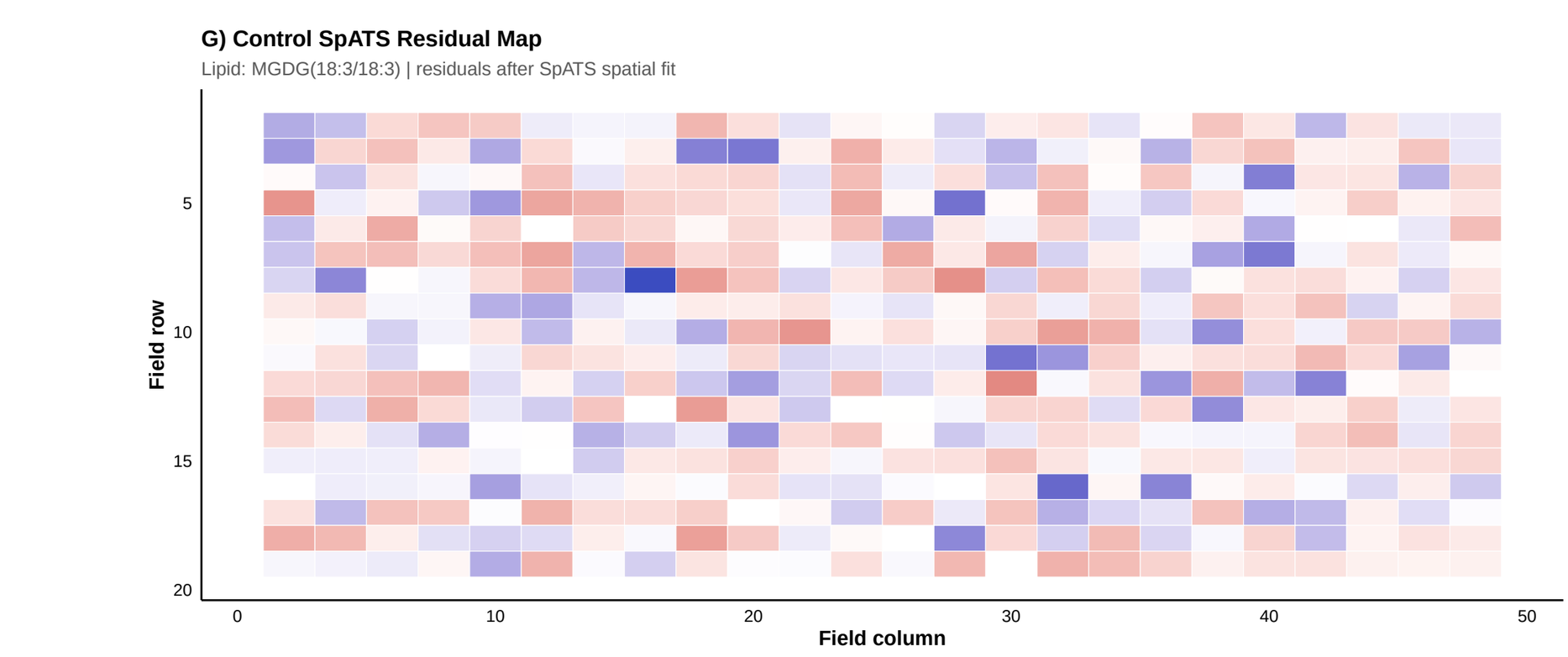

B) LowInput − Run Order TIC
Blank n=289 | Check n=34 | ISTD n=74 | QC n=17 | Sample n=363 | QC TIC CV=6.0%
TIC
Run Order
QC Pre
QC Post
Type Blank Check ISTD QC Sample

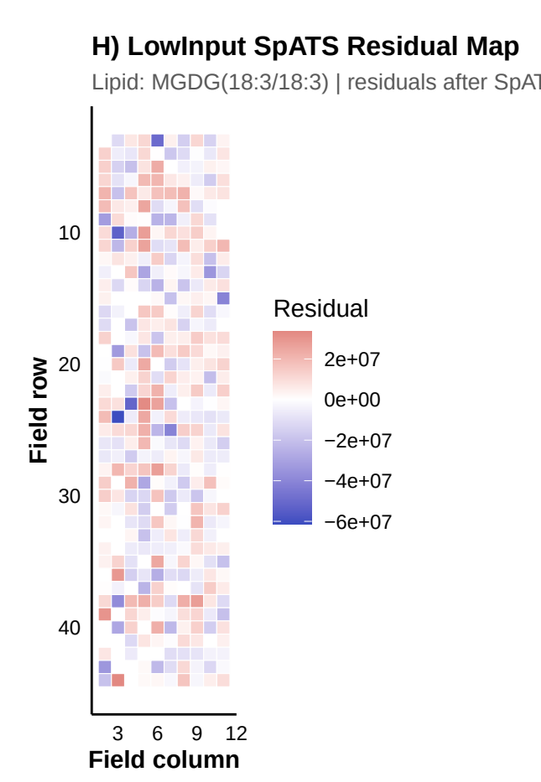

C) Control: QC−RSD After SERRF
QC−RSD < 30%: 75.7% −> 94.1%
Median QC−RSD (%)
3.6%
2.0%
Before SERRF
After SERRF


D) LowInput: QC−RSD After SERRF
QC−RSD < 30%: 77.3% −> 89.3%
Median QC−RSD (%)
1.6%
0.3%
Before SERRF
After SERRF


E) Control PCA: Raw vs SERRF
Matched injections: 524 | matched features: 3244 | dashed ellipse = QC cluster
PC2 (3.1%)
PC1 (78.4%)
Type Check ISTD QC Sample
Stage Raw SERRF


F) LowInput PCA: Raw vs SERRF
Matched injections: 471 | matched features: 3328 | dashed ellipse = QC cluster
PC2 (2.1%)
PC1 (80.7%)
Type Check ISTD QC Sample
Stage Raw SERRF


G) Control SpATS Residual Map
Lipid: MGDG(18:3/18:3) | residuals after SpATS spatial fit
Field row
Field column
Residual
2e+07
0e+00
−2e+07
−4e+07
−6e+07


H) LowInput SpATS Residual Map
Lipid: MGDG(18:3/18:3) | residuals after SpAT
Field row
Field column
Residual
2e+07
0e+00
−2e+07
−4e+07
−6e+07